

QUANTITATIVE APTITUDE CODING & DECODING

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CODING & DECODING

- ✓ Introduction
 - ✓ How to solve questions
 - ✓ The Alphabet concept
 - ✓ Various types of Questions
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INTRODUCTION

- ❑ In questions on coding-decoding, a word (basic word) is coded in a particular way and the candidates are asked to code other word in the same way.
 - ❑ The coding and decoding tests are set up to judge the candidate's ability to decipher the rule that has been followed to code a particular word/message and break the code to decipher the message.
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How to Solve Coding – Decoding Questions

Some of the major types of coding logics are

- ☐ Constant addition in the position of letters.
- ☐ Constant subtraction in the position of letters.
- ☐ Denoting the position of letters in the whole alphabetical order.
- ☐ Addition of the positions of all the letters to make code for the word.
- ☐ Constant addition and subtraction alternatively in the position of all the letters.
- ☐ Square of the number of letters in the word.
- ☐ Arranging the letters in the alphabetical order.
- ☐ Arrangement of letters in the word given in reverse order.
- ☐ Interchanging each pair of the letters, in the given word.
- ☐ Constant addition and then reversal of the letters to form the final word.

TRICKS & TIPS

❑ THE ALPHABET & EJOTY CONCEPT:

- ❑ There are exactly 26 letters in English alphabets.
- ❑ A-Z (1 to 26). A-1, B-2; C-3, and so on.
- ❑ We should learn numbering of every letter.
- ❑ So for better convenience, you can learn two words which help you to remember numbering of letters which are given below.

A	B	C	D	E
F	G	H	I	J
K	L	M	N	O
P	Q	R	S	T
U	V	W	X	Y
		Z		
5	10	15	20	25
E	J	O	T	Y

Reverse EJOTY: When we start count Alphabets from the Z to A, this is called reverse EJOTY.

TYPES OF CODING – DECODING

TYPE 1 – LETTER CODING

- ❑ In this type, the real alphabets in a word are replaced by certain other alphabets according to a specific rule to form its code.
- ❑ You need to detect the common rule and answer the questions accordingly.

Q. If SYSTEM is written as SYSMET and NEARER is written as AENRER, how will FRACTION be written?

A. ACRFTION B. NOITCARE C. CARFNOIT D. TIONCARE

In this question word SYSTEM AND NEARER contains 6 alphabets

Out of these 6 alphabets first three and last three alphabets are written in reverse order,

So code for FRACTION will be

FRAC	TION
CARF	NOIT

Answer: C

TYPE 1 – LETTER CODING

If in a certain language CARROM is coded as BZQQNL, which word will be coded as HOUSE?

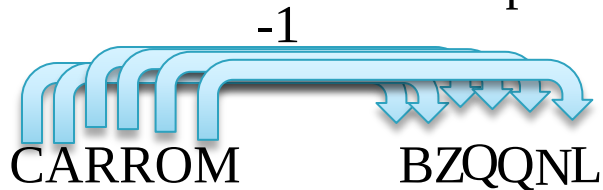
A. GNTRD

B. IPVTF

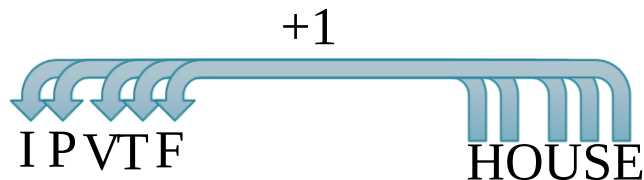
C. ETVPI

D. DRTNG

Each letter of the word is one step ahead of the corresponding letter of the code.



So the word which is coded as HOUSE will be



Answer: B

TYPE 1 – LETTER CODING

Q. If POLISH is written as MLIFPE, then DIG is written as
A. GLJ B. CHI C. ECH D. AFD


POLISH MLIFPE


DIG AFD

Answer: D

TYPE 1 – LETTER CODING

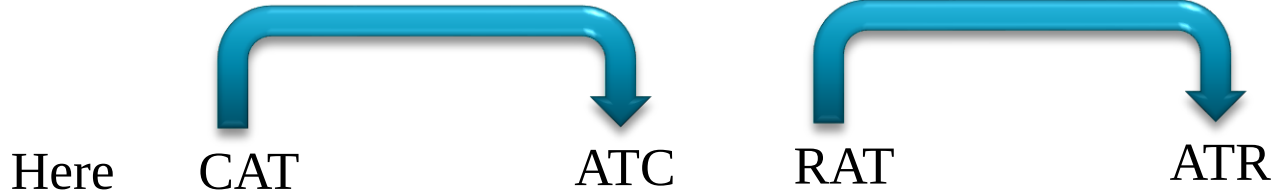
Q. If CAT is written as ATC and RAT is written as ATR, how will MAT be written?

A. AAT

B. MTA

C. TMM

D. ATM



Which means first alphabet is written at last position,
Second alphabet is written at first position
Third alphabet is written at second position.



Answer: D. ATM

TYPE 2 – NUMBER CODING

- ☐ In these questions, either numerical code values are assigned to a word or alphabetical code letters are assigned to the numbers.
- ☐ You need to detect the common rule and answer the questions accordingly.

Q. If DELHI is coded as 73541 and CALCUTTA as 82589662, how can CALICUT be coded?

A. 5279431 B. 5978213 C. 8251896 D. 8543691

Sol:- Since, in CALCUTTA, C is coded as 8, A is coded as 2, L as 5, I as 1 (In DELHI), U as 9, T as 6.

So, code for CALICUT IS 8251896.

Answer : Option C

TYPE 2 – NUMBER CODING

Q. In a certain code, RIPPLE is written as 613382 and LIFE is written as 8192. How is PILLER written in that code ?

A. 318826

B. 318286

C. 618826

D. 338816



Common alphabet is I , whose number code will be 1

And L , whose number code will be 8

Now code for E → 2

P → 3

R → 6

Therefore, code for

PILLER → 318826

Answer : Option A

TYPE 2 – NUMBER CODING

Q. If ROSE is coded as 6821, CHAIR is coded as 73456 and PREACH is coded as 961473, what will be the code for SEARCH ?

A. 246173 B. 214673 C. 214763 D. 216473

ROSE  6821 CHAIR  73456 PREACH  961473

SEARCH  214673

Answer : Option B

TYPE 2 – NUMBER CODING

If in a certain code, TWENTY is written as 863985 and ELEVEN is written as 323039, how is TWELVE written in that code?

- A. 863203 B. 863584 C. 863903 D. 863063

TWENTY → 863985

ELEVEN → 323039

TWELVE → 863203

Answer : Option A

TYPE 3 NUMBER TO LETTER CODING

The number in each question below is to be codified in the following code:

Digit	7	2	1	5	3	9	8	6	4
Letter	W	L	M	S	I	N	D	J	B

1. 184632

A. MDJBSI B. MDJBIL C. MDJBWL D. MDBJIL E. None of these

Answer : Option D

2. 879341

A. DWNIBS B. DWNBIM C. DWNIBM D. NDWBIM E. None of these

Answer : Option C

3. 64928

A. JBNLD B. JBLND C. BJNLD D. DBNLS E. None of these

Answer : Option A

TYPE 4 – MIXED LETTER CODING

Q. If 'ish lto inm' stands for 'neat and tidy' ; 'qpr inm sen' stands for 'small but neat' and 'hsm sen rso' stands for 'good but erratic', what would 'but' stand for ?

A. inm B. qpr C. sen D. hsm

ish lto inm  neat and tidy

'qpr inm  sen  small  but neat

hsm  sen rso  good  but erratic

but  sen

Answer : Option C

TYPE 4 – MIXED LETTER CODING

In a certain code language, 'col tip mot' means 'singing is appreciable' ; 'mot baj min' means 'dancing is good' and 'tip nop baj' means 'singing and dancing' , which of the following means 'good' in that code language ?

A. not B. min C. baj D. Can't be determined E. None of these

col tip **mot** → singing is appreciable
mot **baj** min → **dancing** is good
tip nop **baj** → singing and **dancing**

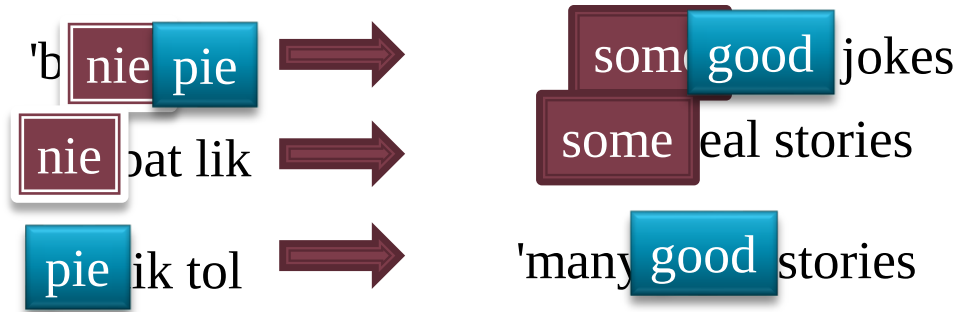
good → min

Answer : Option B

TYPE 4 – MIXED LETTER CODING

In a certain code, 'bi nie pie' means 'some good jokes' : 'nie bat lik' means 'some real stories' ; and 'pie lik tol' means 'many good stories'. Which word in that code means 'jokes' ?

A. bi B. nie C. pie D. Can't be determined E. None of these



jokes → bi

Answer : Option A

Type 5: SUBSTITUTION CODING

In this section an object names are substituted with different object names. We should carefully trace the substitution and answer given question.

If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the color of human blood?

A. Red B. Green C. Blue D. Yellow

Sol. The color of the human blood is 'red'
and as it is given that 'red' is called yellow'.
So, the color of human blood is yellow'

PRACTICE QUESTIONS

If 'diamond' is called 'gold', 'gold' is called 'silver', 'silver' is called 'ruby' and 'ruby' is called 'emerald', which is the cheapest jewel?

- (A) Diamond (B) Silver (C) Gold (D) Ruby (E) Emerald

We know that 'silver' is cheapest.

But as given, 'silver' is called 'ruby'

So, 'ruby' is the cheapest.

Hence, the answer is (d)



PRACTICE QUESTIONS

If 'eye' is called 'hand', 'hand' is called 'mouth', 'mouth' is called 'ear', 'ear' is called 'nose' and 'nose' is called 'tongue', with which of the following would a person hear?

- (A) Eye (B) Mouth (C) Nose (D) Ear (E) Tongue

A person hears with his 'ear'

But as per the given information, 'ear' is called 'nose'.

So, a person will hear with the 'nose'.

Hence, the answer is (C).



PRACTICE QUESTIONS

If 'pen' is 'table', 'table' is 'fan', 'fan' is 'chair' and 'chair' is 'roof', on which of the following will a person sit?

(a) Fan

(b) Chair

(c) Roof

(d) Table

(e) Pen

A person will sit on a 'chair'
and as given, but a 'chair' is called 'roof'.

So, a person will sit on the 'roof'.

Hence, the answer is (C).

Type 6: SYMBOL CODING

In this type of questions either alphabetical code values are assigned to symbols or symbols are assigned to alphabets.

In a certain code 'TOME' is written as '@ \$* ?' and 'ARE' is written as '· £?' How can 'REMOTE' be written in that code?

Sol. From the data we have T = @, O = \$, M = *, E = ? and A = ·
, R = £, E = ?

Hence REMOTE is coded as £ ? * \$ @ ?

PRACTICE QUESTIONS

Study the data given below and answer the following questions:

'Royal Monarch **Regal**' is written as @ # *

Regal 'Legacy Gold' is written as * % ?,

'Hope Gold Life' is written as % & \$,

'Regal Monarch Morals' is written as # * ∇

Q. What will be the code for Regal?

A. % B. * C. \$ D. # E. ?

Ans B *

PRACTICE QUESTIONS

Study the data given below and answer the following questions:

‘Royal Monarch Regal’ is written as @ # *,

‘Regal legacy Gold’ is written as * % ?,

‘Hope Gold Life’ is written as % & \$,

‘Regal Monarch Morals’ is written as # * ∇

What will be the code for “Gold Legacy”?

A. # & B. % # C. % ? D. * ∇ E. & \$

Ans C % ?

PRACTICE QUESTIONS

Study the data given below and answer the following questions:

‘Royal **Monarch** Regal’ is written as @ # *,

‘Regal legacy Gold’ is written as * % ?,

‘Hope Gold Life’ is written as % & \$,

‘Regal **Monarch** Morals’ is written as # * ∇

Which word is coded as #?

A. Moral

B. Life

C. Regal

D. Monarch

E.

Legacy

Ans D **Monarch**

PRACTICE QUESTIONS

In a certain code, RAID is written as %#*\$, RIPE is written as % * @ ©. How is DEAR written up in that code?

(A) \$ * 37

(B) 3\$7*

(C) \$©#*

(D) \$©#%

According to the question,

Letters

Symbol code

Hence

R	A	I	D	R	I	P	E
%	#	*	\$	%	*	@	©

D	E	A	R
\$	©	#	%

Ans D

PRACTICE QUESTIONS

In a code language, TATA is written as 400, then what should we write for PAPA in the same language?

(A) 32

(B) 1616

(C) 800

(D) 256

$$\begin{array}{ccccccc} \mathbf{T} & & \mathbf{A} & & \mathbf{T} & & \mathbf{A} \\ 20 & \times & 1 & \times & 20 & \times & 1 & = & 400 \end{array}$$

$$\begin{array}{ccccccc} \mathbf{P} & \times & \mathbf{A} & \times & \mathbf{P} & \times & \mathbf{A} \\ 16 & \times & 1 & \times & 16 & \times & 1 & = & 256 \end{array}$$

Ans D

PRACTICE QUESTIONS

In a code language, PEN is written as 1517461315, then what should be written for COPY in the same code language?

(A) 23141516172526

(B) 23141615172526

(C) 23411615172425

(D) 24141615172426

P	E	N	C	O	P	Y
16	5	14	3	15	16	25
15	17	4	6	13	15	26

Ans D

PRACTICE QUESTIONS

If EARTH is written as FCUXM in a certain code. How is MOON written in that code?

(A) NQRT

(B) NRQT

(C) NQRR

(D) RRQN

E	A	R	T	H
+1	+2	+3	+4	+5
↓	↓	↓	↓	↓
F	C	U	X	M

M	O	O	N
+1	+2	+3	+4
↓	↓	↓	↓
N	Q	R	R

Ans D

PRACTICE QUESTIONS

If DELHI is written as EDMGJ in a certain code. How is NEPAL written in that code?

(A) ODQZM

(B) ODQMZ

(C) MDOZK

(D) MDOKZ

D	E	L	H	I
+1	-1	+1	-1	+1
↓	↓	↓	↓	↓
E	D	M	G	J

N	E	P	A	L
+1	-1	+1	-1	+1
↓	↓	↓	↓	↓
O	D	Q	Z	M

Ans A

PRACTICE QUESTIONS

In a certain code, HARYANA is written as 8197151, how is DELHI written in that code?

We used the number of alphabets here.

$$H = 8$$

$$A = 1$$

$$R = 18 = 1+8 = 9$$

$$Y = 25 = 2+5 = 7$$

HARYANA 8197151

For DELHI

$$D = 4$$

$$E = 5$$

$$L = 12 = 1+2 = 3$$

$$H = 8$$

$$I = 9$$

Hence, DELHI is written as 45389.

PRACTICE QUESTIONS

In a certain code, NEWYORK is written as 111, how is NEWJERSEY written in that code?

In NEWYORK

$$N = 14, \quad E = 5, \quad W = 23, \quad Y = 25, \quad O = 15, \quad R = 18, \quad K = 11$$

$$\text{Total} = 14 + 5 + 23 + 25 + 15 + 18 + 11 = 111$$

In NEWJERSEY

$$\begin{aligned} \text{Total} &= 14 + 5 + 23 + 10 + 5 + 18 + 19 + 5 + 25 \\ &= 124 \end{aligned}$$

1. In a certain code language 'SURVEY' is coded as 'TWUZJE' and 'APPLY' is coded as 'BRSPD'. Then how does 'WISDOM' be coded in that language?

A. XKTHTS

B. XKVHTS

C. YLISUV

D. YLISVU



2 In a certain code language, 'MOTIVATION' is written as 'VITOMNOITA' and 'DECODING' is written as 'OCEDGNID'. Then how 'FACEBOOK' will be written in that language?

A. KOOBFACE B. KOOBECAF C. ECAFKOOB D. ECAFBOOK



3. In a certain code language, '*lily*' is called '*rose*', '*rose*' is called '*gladiola*', '*gladiola*' is called '*lotus*' and '*lotus*' is called '*poppy*' and '*poppy*' is called '*sunflower*'. Then which is the national flower of India?

A. sunflower

B. poppy

C. rose

D. gladiola

4. In a certain code language, all the animals which can live in water are called *snakes* and those which fly in sky are called *swimmers* and those which can walk are called *hunters* and those animals who crawl are called *flying*. Then what will a *tortoise* be called?

A. swimmers

B. snakes

C. hunters

D. flying



5. In a certain code language, '*kings*' are '*queens*', '*queens*' are '*aces*', '*aces*' are '*jacks*', '*jacks*' are '*spades*', '*spades*' are '*diamonds*'. '*diamonds*' are called '*kings*'. Then who is the lady supreme in the kingdom?

A. aces

B. diamonds

C. kings

D. queens



6. In a certain code language, 'CHILD' is coded as '32546', 'MODEL' is coded as '17684', then 'OLEIC' is coded as?

A. 74213

B. 45321

C. 77853

D. 74853

7. If in a certain code language, '186' is coded as '*I love India*', '983' is coded as '*India has culture*', '346' is coded as '*we love culture*'. Then the code for 'we' and 'I' is.
- A. 9, 1 B. 4, 8 C. 9, 8 D. 4, 1
- 

8. In a certain code language, '423' is coded as '*Hyderabad is beautiful*', '735' is coded as '*Beautiful flower pleasant*', '524' is coded as '*Hyderabad is pleasant*'. Then the code for '*beautiful*' and '*pleasant*' are?

A. 5, 3 B. 3, 5 C. 5, 2 D. 5, 4



9. '456' means '*she is beautiful*'. '632' means '*he is handsome*'. '517' means '*she works hard*', '486' means '*flower is beautiful*'. Then codes of '*flower*' and '*is*' are?

A. 4, 5

B. 2, 7

C. 8, 6

D. 6, 7

10. If '*RELATION*' is written as '*OFMBUJPS*'. Then how '*IMPORTANT*' is written?

A. JNQPSUBOU B. FNQPSUBOU C. UNQPSUBOJ D. FNQPSUBOR

11. If '*ORKUT*' is coded as '*LIPFG*'. How can '*TUMBLR*' be coded?
- A. RBLMUT B. GFNYOI C. IOYNFG D. RBLMUT

12. In a certain language, '*DIRECTOR*' is coded as '*IDERTCRO*', then how '*CHAMPION*' is coded in that language?

A. HACMNOPI B. HCMNOPIA C. HCMAIPON D. HCMAIPNO



13. If 'MOTIVATION' is coded as 'NOUIWAUIPN' then 'STRONG' is coded as?
A. TTSOOG B. TTRSMG C. SSTTMG D. GMOSTT



14. If 'SURVEY' is coded as 'TTSUFX' then 'WISDOM' is coded as?
A. XHTCPL B. XHTEPL C. XHRCPL D. XHTCNL



15. Which of the following words correctly decode the word 'TBIZLJB' if the simple alphabet shifting code is used?

A. ARTISTS B. FORWARD C. WELCOME D. BLBYAZI



In a certain code language:

'ju tu ne' means 'rain is heavy'

'paa ble ka' means 'winter was hot'

'sue tu qle' means 'rain drops fall'

'ka ne da' means 'summer is hot'

'tho pic joo' means 'clever student reads'

'ble qle tho' means 'winter drops clever'

16. What is the code for 'was'?

A. ble B. paa C. ka D. None

17. What is the code for 'fall'?

A. slu B. qle C. tho D. tu

18. What is the code for 'reads'?

A. pic B. joo C. tho D. 'pic' or 'joo'

19. What is the code for 'winter heavy rain'?

A. ble ka ju B. ju tue qle C. ju tu ble D. sue tu pic

20. What is the code for 'summer is clever'?

A. da tho ne B. pic sue ble C. ble tu ne D. ble sue ne

In a certain code language:

'7462' means 'you are very intelligent'

'491' means 'you can see'

'896' means 'is can are'

'2357' means 'they seem very intelligent'

'0852' means 'how intelligent they is'

21. What is the code for 'see'?

A. 1 B. 9 C. 4 D. 6

22. What will be the decode of the digit '0'?

A. they B. how C. is D. intelligent

23. What will be the decode of the digit '2'?

A. seem B. they C. intelligent D. very

24. What will be the code for 'they can see'?

A. 915 B. 916 C. 718 D. 719

25. What will be the code for 'how very is'?

A. 714 B. 836 C. 807 D. cannot be determined

In a certain code language: BAT is written as AZS and ROOM is written as QNNL.

26. The code for HELMET is?

- A. DKGSLD B. DGKLDS C. GDKLDS D. XAMBX Y**

27. The code for PRAXIS is?

- A. RHWZQO B. QZWOHR C. PQRSTU D. OQZWHR**

28. The code for DAMAGE is?

- A. CZZLFD B. DCZFLZ C. XBPBAL D. CZLZFD**

29. The code for SACHIN is?

- A. MHGBZR B. RZBGHM C. RZBHGM D. RZBMHG**

30. The code for LAPTOP is?

- A. KZOONS B. ONOSKZ C. KZOSNO D. ONSOZK**
- 

DIRECTION SENSE

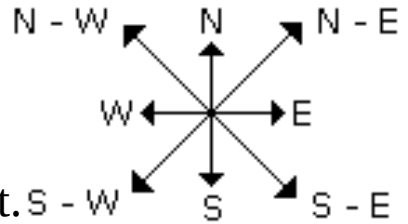
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DIRECTION

Direction is the information contained in the relative position of one point with respect to another point without the distance information.

- ❖ Four main directions - East, West, North and South.
- ❖ There are four cardinal directions - North-East (N-E), North-West (N-W), South-East (S-E), and South-West (S-W)
- ❖ At the time of sunrise if a man stands facing the east, his shadow will be towards west.
- ❖ At the time of sunset the shadow of an object is always in the east.
- ❖ If a man stands facing the North, at the time of sunrise his shadow will be towards his left and at the time of sunset it will be towards his right.
- ❖ At 12:00 noon, the rays of the sun are vertically downward hence there will be no shadow.



1. Deepak starts early in the morning from his house and jogs towards north for 2 km. Then he turns left and jogs for 1 km.

Now, what is his position with respect to his house? _____

Which direction is he facing? _____

He then takes a right turn and jogs 1 km.

Now his position with respect to his house is _____

The direction that he is facing is _____

The distance of his current position from his house is _____



2. A man walks 10 km towards north. From there he walks 6 km towards south and then 3 km east. How far and in which direction is he with reference to his starting point?

3. You leave home, walk one kilometer south, then one kilometer east and finally one kilometer north. If you are home again at the end of your walk, where is your home?



If you start running from a point towards north and after covering 4 km, you turn to your left and run 5 km and then again turn left and run another 6 km and before finishing you take a left turn and run for 1 km.

- 4. How far are you from the Starting point?
A. $1\sqrt{5}$ B. $2\sqrt{5}$ C. $3\sqrt{5}$ D. $4\sqrt{5}$**
 - 5. In which direction will you be at the end (with reference to the starting point)?
A. East B. West C. North D. South**
 - 6. After taking the second turn, in which direction will you be running?
A. East B. West C. South D. North**
- 

7. A child is looking for his father. He went 90 m east before turning right. He then went 20 m before turning to the right again to look at his uncle's place 30 m from that point. His father was not there. From there he went 100 m north, before meeting his father in a street. How far was the father from the starting point?
- A. 80 B. 100 C. 140 D. 260
- 

8. Mr. Mirza Baig travels 7 km north, then turns right and walk 3 km. He again turns to his right and moves 7 km forward. How many km, is Mirza Baig away from the place of his starting the journey ?

A. 7 B. 3 C. 6 D. 14



9. Mr. Naveen & Ms. Reeta both start from a point towards North. Naveen turns left after walking 10 km. Ms. Reeta turns right after walking the same distance. Naveen waits for some time and then walks another 5 km, whereas Reeta walks slowly for 5 km after turning right. They both turn to the south and walk 15 km forward. How far is Naveen from Reeta?

A. 15

B. 10

C. 8

D. 12



10. The door of Vinod's house faces east. From the backside of his house, he walks 50 m towards west, then turns right and walks 50 m. Finally, he turns left and stops walking after 25 m. Now in which direction is he from the starting point?

A. South-East B. North-East C. South-West D. North-West



Radhika walks 2 km north and turns right and walks 4 km. He then makes a second right turn and walks 4 km. He again turns right and travels 4 km. Here she meets Rashmi coming from the opposite direction and they both stop.

- 11. If the starting point is marked as A and the finishing point is marked as B, then what is the distance between these points?**
A. 10 B. 8 C. 6 D. 2
- 12. Towards which direction was Radhika coming after the third turn?**
A. East B. West C. South D. North
- 13. After taking the second turn, in what direction was Radhika walking?**
A. East B. West C. North D. South
- 14. If Radhika is to again reach the point from where she started, then in which direction will she have to walk?**
A. East B. South-East C. North D. North-East
- 

Answer the following questions based on the information given below.

- **There are 6 check posts A, B, C, D, E & F.**
- **Check post F is 15 km to north of D, which is 25 km to the North-East of B.**
- **Check post A is 5 km West of E and 15 km to the South-West of C.**
- **B, A and E are in a straight line and in the same sequence.**
- **The Check Post B and E are 30 km apart.**

15. Which check post is the farthest to the southwest of D?

- A. A B. B C. C D. D**

16. Which post is the nearest and to the north-East of E?

- A. A B. B C. C D. D**

17. If a Jeep moves from E to F via A, B and D, how much distance will it have to cover (in km)?

- A. 70 B. 120 C. 100 D. 90**

- 18. A man walks 8 Km towards North. From there he turns right and walks 7 Km. Again he turns right and walks 8 Km. Now he turns left and walks 10 Km. Again he turns left and walks 8 Km. Now he turns right and walks 9 Km. Again he turns right and walks 8 Km. Find the distance and direction he is from the starting point?**
- A. 30 Km, West B. 44 Km, East C. 26 Km, East D. 30 Km, East**
- 

19. Aman walks 2 Km towards West, then he turns 3 Km of to his left. From there he walks 10 Km to his right. Finally he walks 2 Km to his left and stops there. How far he is from the starting point?

A. 17 Km

B. 13 Km

C. 15 km

D. 11 Km



20. After walking 8 Km, Sachin turned to the right and then walked 6 Km. After that he turned to his left and walked 12 Km. In the end, he was moving towards the North. From which direction did Sachin started his journey?

A. East

B. West

C. North

D. South



21. For a competitive exam only 4 members attended. The examiner made them to sit in such a way that P sits West of Q and South of R, who sits East of S. Determine the direction of S from Q.

A. North

B. North-West

C. North-East

D. West



22. 5 Poles P, Q, R, S and T are located in such a way that T is to the East of Q. Q is to the South of R. P is to the West of Q. S is to the North of P and $\angle RSP$ is equals to 90° . Then find the relative position of S from R?

A. North

B. South

C. East

D. West



23. 6 poles A, B, C, D, E and F are located in such a way that B is to the North of A and to the West of E. C is to the South of E and to the North of D. A is to the West of C. F is to the East of both B and E and North-East to the C. In which direction of B is D located?

- A. South-East B. South-West C. North-East D. South**

24. If East becomes North-West, North-West becomes South, South becomes North-East and so on. Then what will North becomes?

- A. North-East B. South-East C. South-West D. North**

**25. If South becomes East, East becomes North and North becomes West and so on.
What will West become?**

A. North

B. South

C. North-East

D. South-East



26. If North-East becomes North-West, North becomes West, South-East becomes South-West and West becomes South and so on. Then what North-West will become?

A. North-East

B. North

C. East

D. South-East



27. One morning Kavitha and Mamatha were talking to each other face to face at a crossing. If Kavitha's shadow was exactly to the right of Mamatha, in which direction was Mamatha facing?

A. East

B. North-East

C. South

D. South-West



28. One morning after sunrise, Raju stand between two poles namely P1 and P2. Raju is facing P1 and he turned 180° clock wise and his face is exactly opposite to P2. He observed that the shadow of P2 is exactly falling to his right. In which direction does P2 is located with respect to Raju?

A. East B. West C. North D. South



29. One day Charan is going to school. He started walking in North direction, after walking some distance, he turned 45° clockwise, then 180° in the same direction. From there he turned 270° in anticlockwise direction. Finally in which direction is Charan facing?

A. North-East

B. North-West

C. North

D. East



30. Kavitha is facing North-East, from there she turned 45° in the clockwise, then she turned 90° in the same direction. Now she turned 135° in the opposite direction of her present direction, then she turned 180° in the opposite direction of her present direction. Then she turned 90° in the anticlockwise direction and again she turned 90° in the same direction. Find her final direction in which she is facing?

A. North-East

B. North-West

C. North

D. East



THANK YOU

